Doppler Tomography using Total Variation Minimization (DTTVM) for X-ray Spectra: User Guide

Naoto Sameshima

October 22, 2026

This code extends DTTVM (Uemura et al., 2015) to X-ray emission lines. It models multiple, possibly overlapping lines using a common velocity map and fixed intensity ratios. Instrumental Gaussian broadening and natural Lorentzian broadening are included through a Voigt profile. The Voigt function is evaluated using the approximation of Smith et al. (2015). The extension only changes the spectral response, retaining the original TV minimization and cross-validation procedures. For the basic DTTVM workflow, see the original user guide.

1 Parameters

The Doppler tomography command is:

```
./dttvm_xray data_file nbin resol lambda lines Eref [Vgamma [thresh]]
```

The cross-validation command is:

```
./dttvm_xray_cv data_file nbin resol lambda_min lambda_max nlambda lines Eref [Vgamma [thresh]]
```

`data_file` Input spectrum file containing radial velocity in $\mathrm{km\,s^{-1}}$, continuum-subtracted flux, and orbital phase.

`nbin` Number of bins along each side of the velocity map. The total number of pixels is `nbin`^{2}.

`resol` Instrumental Gaussian standard deviation in $\mathrm{km\,s^{-1}}$.

`lambda` Weight of the total variation term. For cross-validation, use `lambda_min`, `lambda_max`, and `nlambda`: the positive lower and upper limits and the number of logarithmically spaced trial values.

`lines` Line-list filename. The format is described below.

`Eref` Reference energy in keV used to define the input velocity coordinate:

$$v = c\left(\frac{E_{\mathrm{ref}}}{E} - 1\right).$$

Positive velocity denotes redshift. Use the same reference energy when preparing the spectrum and running the code.

`Vgamma` Optional systemic velocity in $\mathrm{km\,s^{-1}}$, subtracted from the input velocities. Default: 0.

`thresh` Optional convergence tolerance for the relative change in the objective function. Default: 10^{-4}.

2 Input file

The `lines` file is a plain-text file with one line component per row and three whitespace-separated columns:

1. Rest energy in keV (> 0).
2. Relative line intensity (≥ 0).
3. Natural Lorentzian FWHM in eV (≥ 0).

The weights are normalized internally by their sum, which must be positive. For example, weights of 2 and 1 specify an intensity ratio of $2 : 1$. All components share the same velocity map, and this ratio is fixed throughout the map. An illustrative two-line file is:

```
# Energy [keV]  Weight  Natural FWHM [eV]
6.400           2.0     0.807
6.387           1.0     1.244
```

Blank lines and comments beginning with `#` are allowed. Setting the natural FWHM to zero gives a Gaussian instrumental profile.

References

Aaron Smith, Chalence Safranek-Shrader, Volker Bromm, and Miloš Milosavljević. The Lyman α signature of the first galaxies. *Mon Not R Astron Soc*, 449(4):4336–4362, June 2015. ISSN 0035-8711. doi: 10.1093/mnras/stv565. URL `https://doi.org/10.1093/mnras/stv565`.

Makoto Uemura, Taichi Kato, Daisaku Nogami, and Ronald Mennickent. Doppler tomography by total variation minimization. *PASJ*, 67(2):22, April 2015. ISSN 0004-6264. doi: 10.1093/pasj/psu154. URL `https://doi.org/10.1093/pasj/psu154`.